# Last time: XSS

CS 161 Spring 2026 - Lecture 20

# Cross-Site Scripting (XSS)

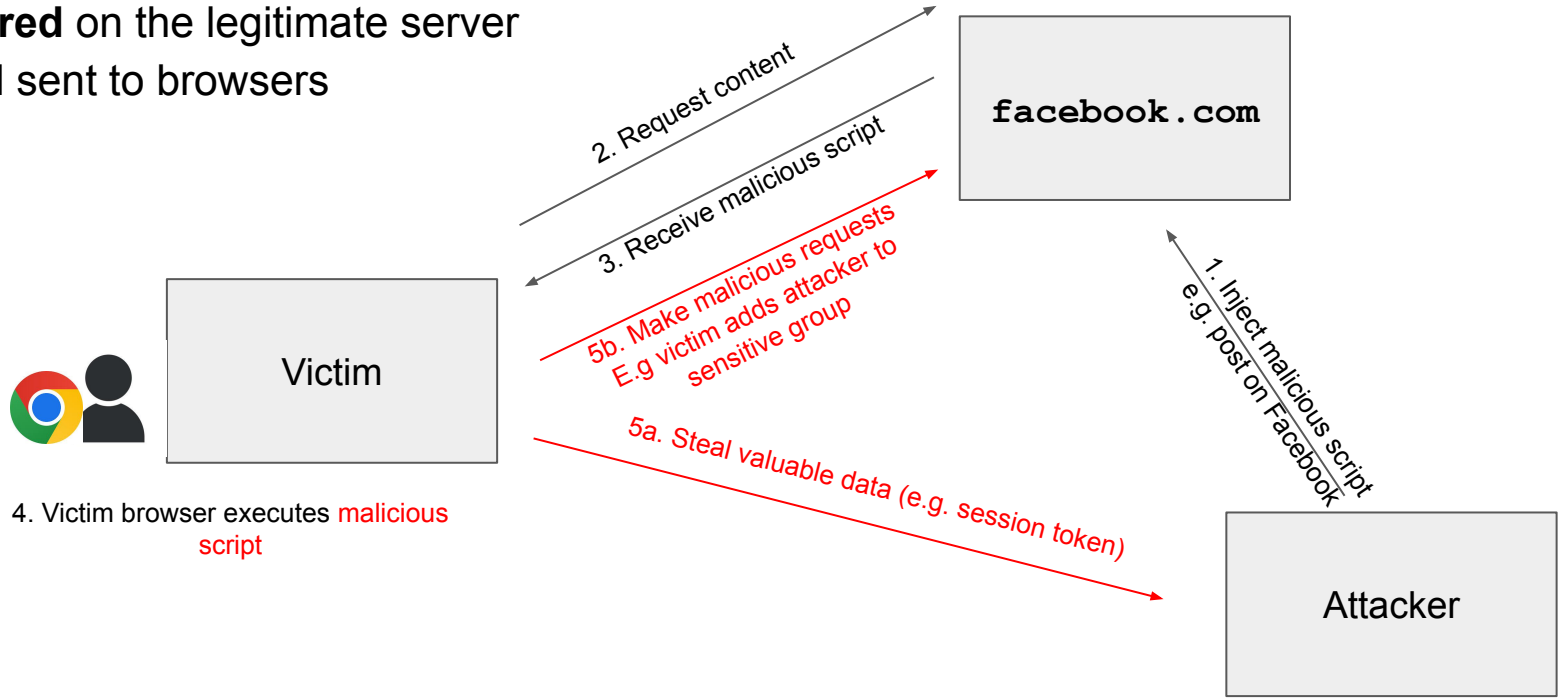


- **An attack in which** the attacker adds malicious JavaScript to a legitimate website that is then viewed by victim users in their browsers
- Two main types of XSS
  - Stored XSS
  - Reflected XSS

# Stored XSS



The attacker's JavaScript is **stored** on the legitimate server and sent to browsers



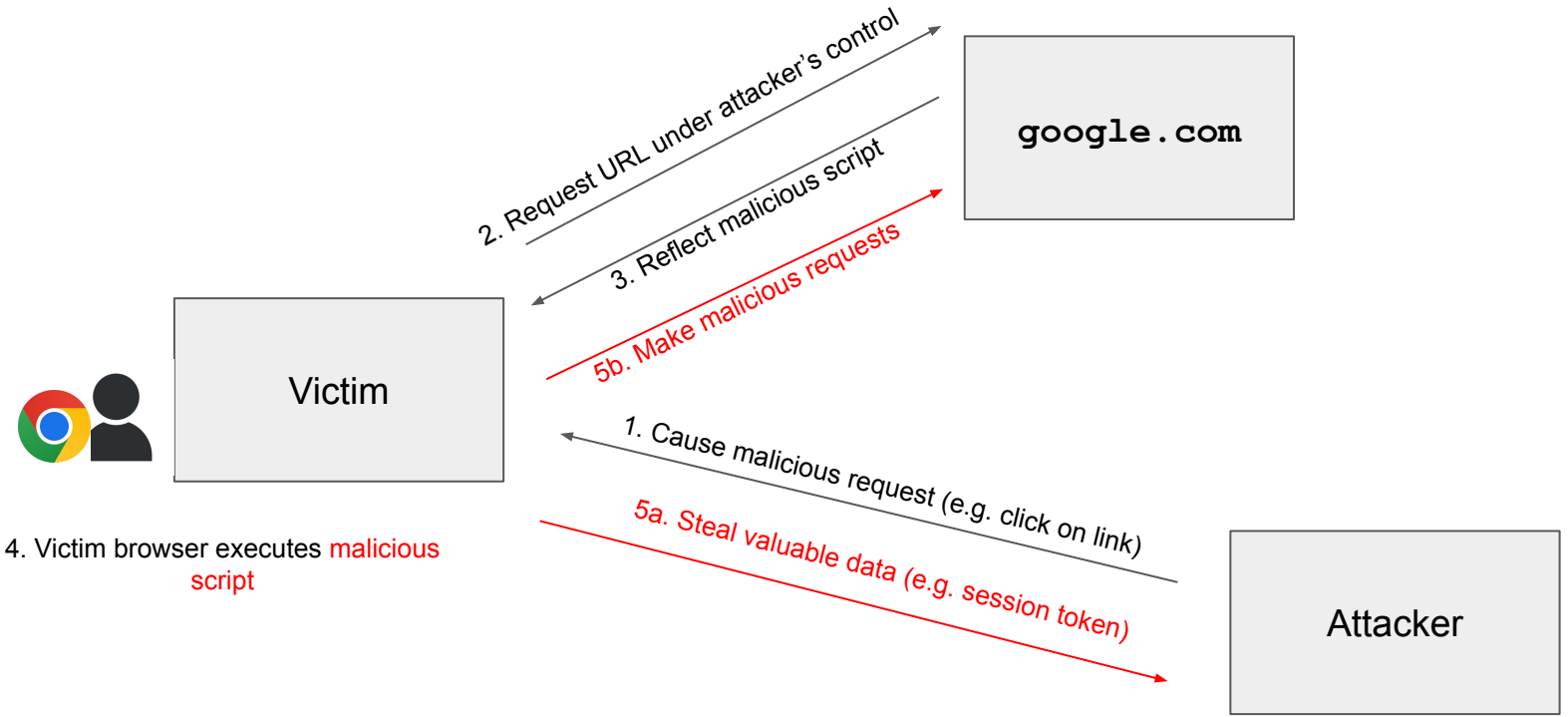
# Reflected XSS



- **Reflected XSS**: The attacker causes the victim to input JavaScript into a request, and the content is **reflected** (copied) in the response from the server
- Classic example: Search
  - If you make a request to `http://google.com/search?q=evanbot`, the response will say "10,000 results for evanbot"
  - If you make a request to `http://google.com/search?q=<script>alert(1)</script>`, the response will say "10,000 results for `<script>alert(1)</script>`"
- Reflected XSS requires the victim to make a request with injected JavaScript

# Reflected XSS

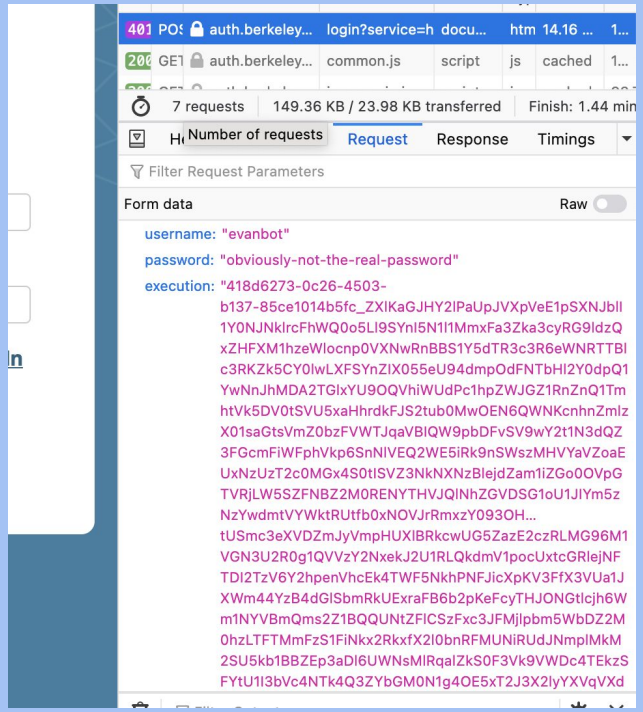




# XSS in the Wild… On CalNet?!



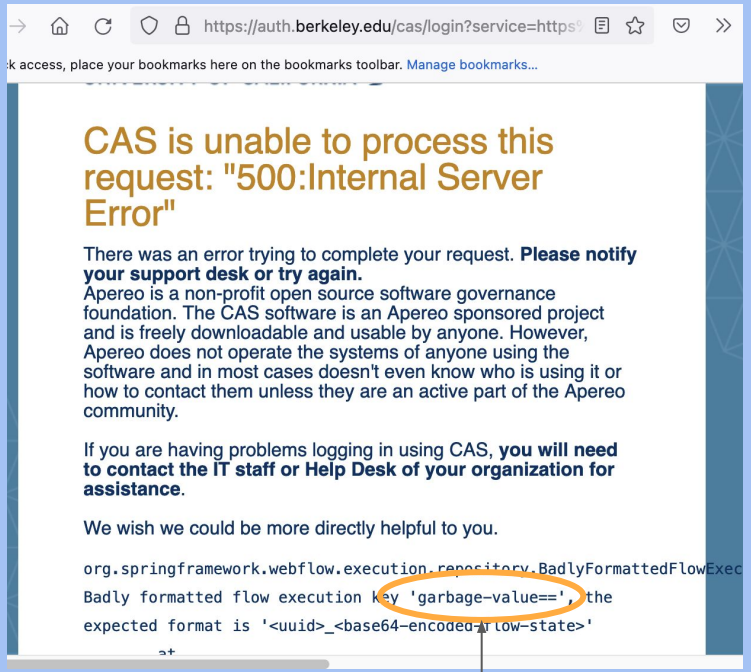
- In 2021, 61C student Rohan Mathur was exploring the CalNet login page
- Fields submitted when logging in:
  - **`username`**: What user am I logging in as?
  - **`password`**: What's the user's password?
  - **`execution`**: Server-side state (like a CSRF token with extra data)



# XSS in the Wild… On CalNet?!



- The server expects **`execution`** to be in a specific format that contains its server-side state
  - What if you muck with **`execution`**?
- The “corrupted” execution key is placed into the error message in the HTML to aid debugging
  - … and CalNet at the time did not escape the execution key



# Constructing an Attack on CalNet



Attack: Force a POST request to CalNet!

```
<html>
  <head>
    <script>
      // When the malicious page finishes loading, automatically submit the form!
      document.addEventListener('DOMContentLoaded', () => {
        document.getElementById('form').submit();
      });
    </script>
  </head>
  <body>
    <!-- Malicious form containing our malicious execution data. -->
    <form id="form" action="https://auth.berkeley.edu/cas/login" method="POST">
      <input name="username" type="text" value="evanbot" />
      <input name="password" type="text" value="obviously-not-the-real-password" />
      <input name="execution" type="text" value="<script>alert('XSS!')</script>" />
    </form>
  </body>
</html>
```

# So What Happened?



- CalNet also accepts **`execution`** parameters over URL query parameters
  - A link like **`https://auth.berkeley.edu/cas/login?execution=<script>alert('XSS!')</script>`** would result in the same attack
- It would only fire if you clicked the button to show the error
- … But if clicked, the injected JavaScript could
  - Present a fake login prompt
  - Steal your CalNet password
  - Log you in as the attacker's account
  - Steal your authentication session token
- Root cause: A >5 year old sample page modified by CalNet
  - CalNet uses software from Apereo, which comes with sample pages for logging in
  - Apereo fixed that bug many years before, but sample pages don't get included in bug fixes!

# Recall: XSS Defenses



- Defense: **HTML sanitization**
  - Idea: Certain characters are special, so server should remove those characters or encode them in a way that will render them harmless
- Escaping: e.g., replace < with `<` with " with `"`
- Rely on trusted libraries to do this for you
- **Content Security Policy** (**CSP**)

```
<html>
<body>
Hello &lt;script&gt;alert(1)&lt;/script&gt;!
</body>
</html>
```

# UI Attacks

Textbook Chapter 23

# User Interface (UI) Attacks



- General theme: The attacker tricks the victim into thinking they are taking an **intended** action, when they are actually taking a **malicious** action
  - Takes advantage of **user interfaces**: The trusted path between the user and the computer
    - Browser disallows the website itself to interact across origins (same-origin policy), but trusts the user to do whatever they want
  - Remember: Consider human factors!
- Two main types of UI attacks
  - Clickjacking: Trick the victim into clicking on something from the attacker
  - Phishing: Trick the victim into sending the attacker personal information

# Clickjacking



- **Clickjacking**: Trick the victim into clicking on something from the attacker
- Main vulnerability: the browser trusts the user's clicks
  - When the user clicks on something, the browser assumes the user intended to click there
- Why steal clicks?
- Why steal keystrokes?
- Discuss with your neighbor!

# Clickjacking



- **Clickjacking**: Trick the victim into clicking on something from the attacker
- Main vulnerability: the browser trusts the user's clicks
  - When the user clicks on something, the browser assumes the user intended to click there
- Why steal clicks?
  - Download a malicious program
  - Like a Facebook page/YouTube video
  - Delete an online account
- Why steal keystrokes?
  - Steal passwords
  - Steal credit card numbers
  - Steal personal info

# Clickjacking: Download buttons

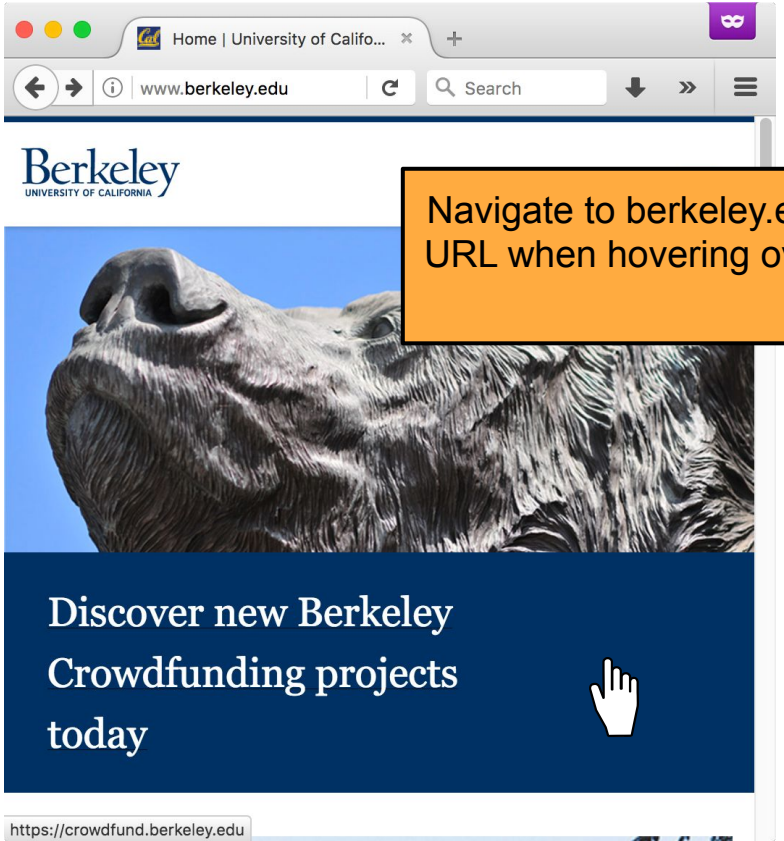


- Which is the real download button?
- What if the user clicks the wrong one?



# Clickjacking





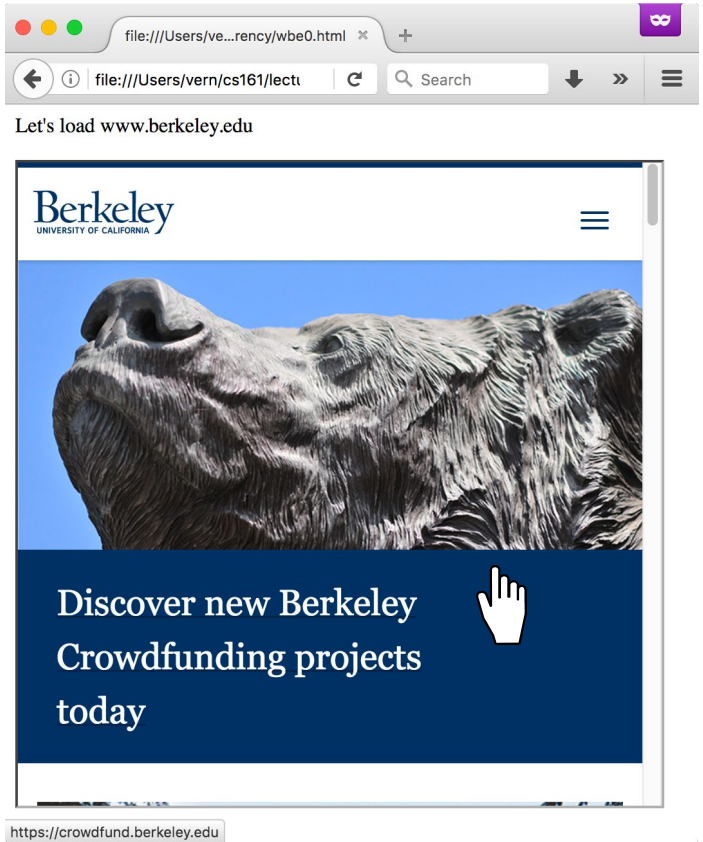
# Clickjacking



Load berkeley.edu in an iframe

Outer frame can't generate clicks because of SOP, but the user can still click…

```
<iframe style="opacity: 1.0"
src="https://www.berkeley.edu/"></iframe>
```

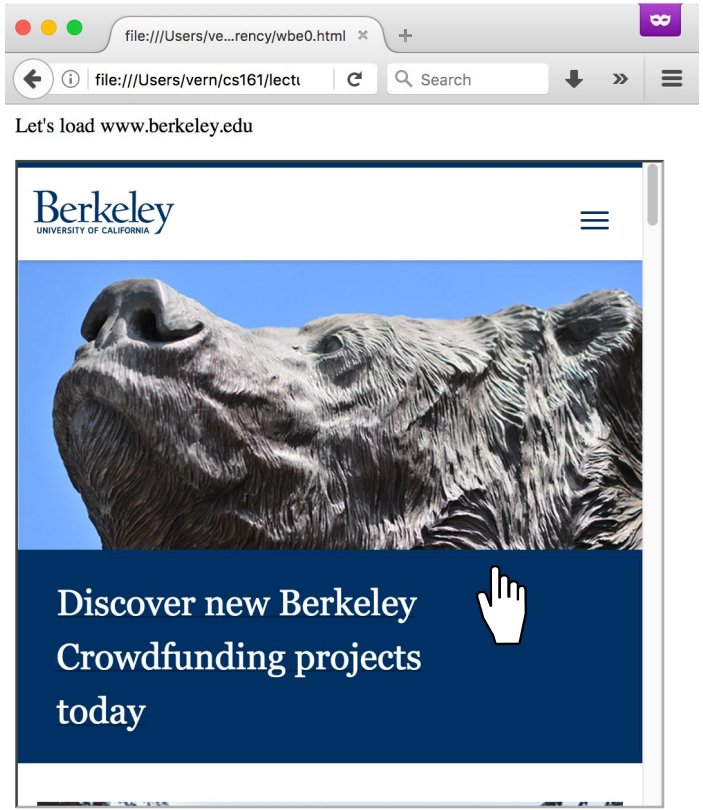


# Clickjacking



Place some enticing content underneath

```
<iframe style="opacity: 1.0"
src="https://www.berkeley.edu/"></iframe>
<p style="margin-top: 210pt"><em>You <b>Know</b>
You Want To Click Here!</em></p>
```

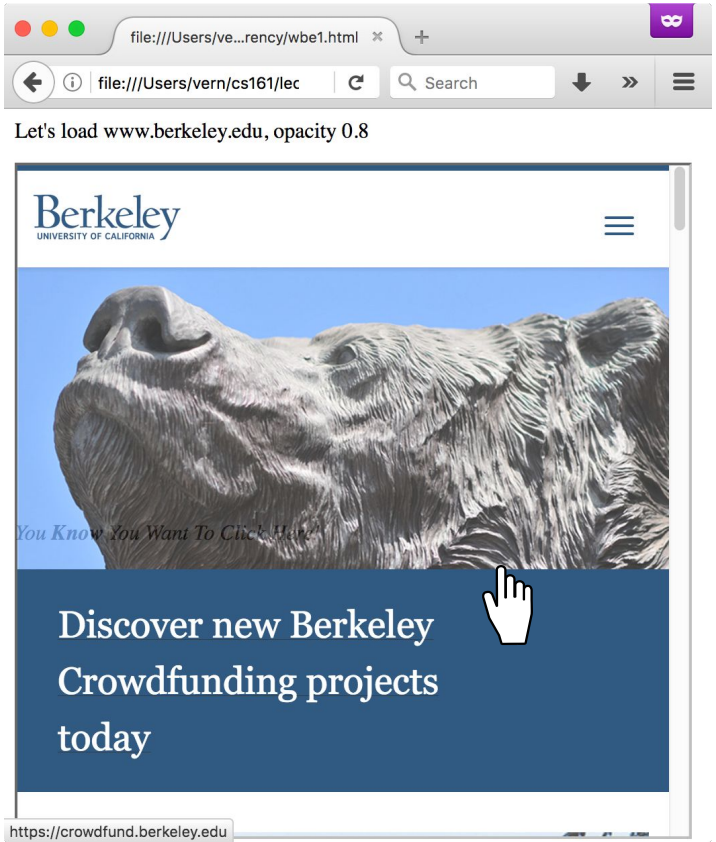


# Clickjacking



Make the iframe slightly transparent…

```
<iframe style="opacity: 0.8"
src="https://www.berkeley.edu/"></iframe>
<p style="margin-top: 210pt"><em>You <b>Know</b>
You Want To Click Here!</em></p>
```

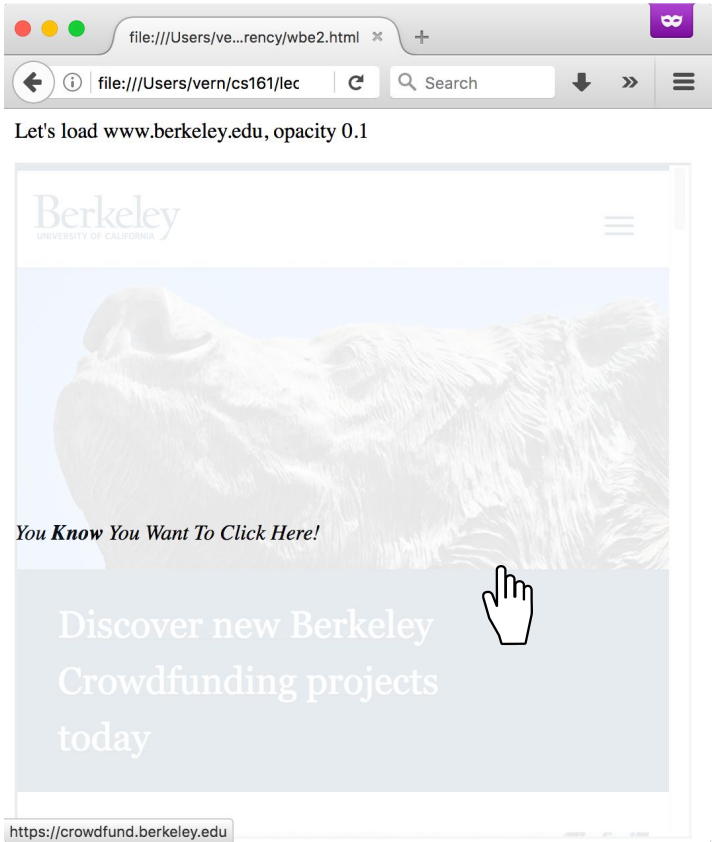


# Clickjacking



Make it *more* transparent

```
<iframe style="opacity: 0.1"
src="https://www.berkeley.edu/"></iframe>
<p style="margin-top: 210pt"><em>You <b>Know</b>
You Want To Click Here!</em></p>
```



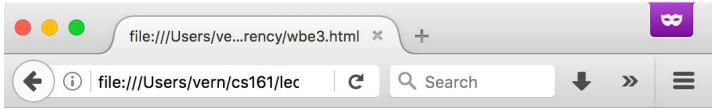
# Clickjacking



Make it *entirely* transparent

But the user still clicks on the iframe!

```
<iframe style="opacity: 0"
src="https://www.berkeley.edu/"></iframe>
<p style="margin-top: 210pt"><em>You <b>Know</b>
You Want To Click Here!</em></p>
```




*You **Know** You Want To Click Here!*

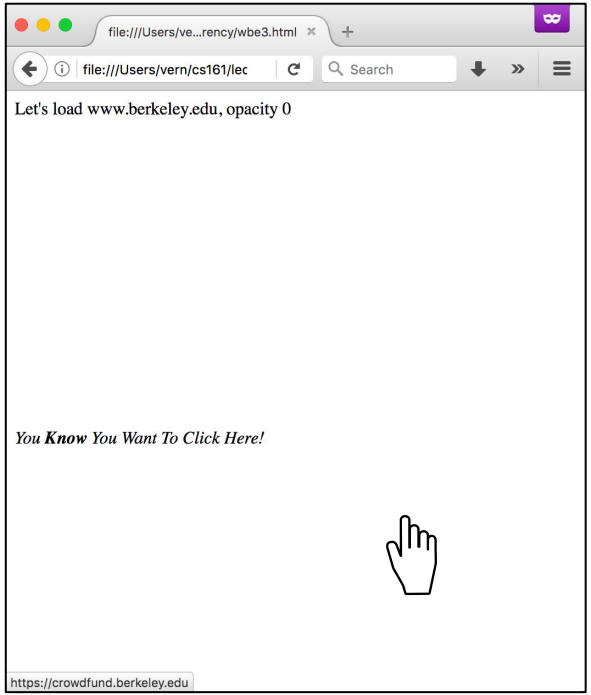


https://crowdfund.berkeley.edu

# Clickjacking: Invisible iframes



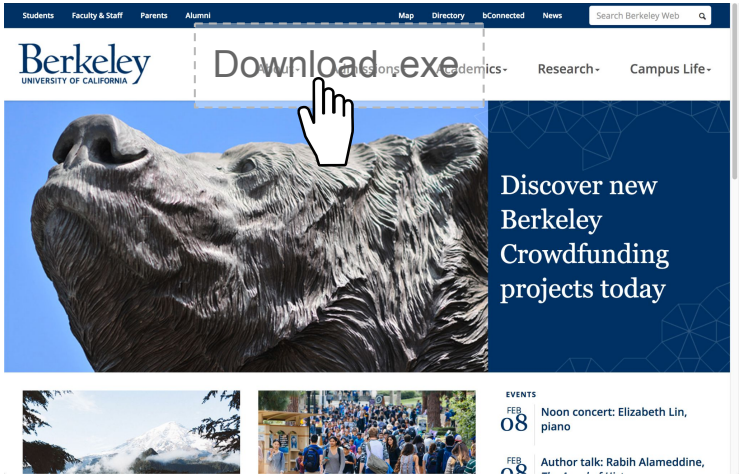
- Variant #1: Frame the legitimate site invisibly, *over* visible, enticing content
  - Victim thinks they're clicking on the attacker's enticing website
  - But their click actually happened on the legitimate website!



# Clickjacking: Invisible iframes



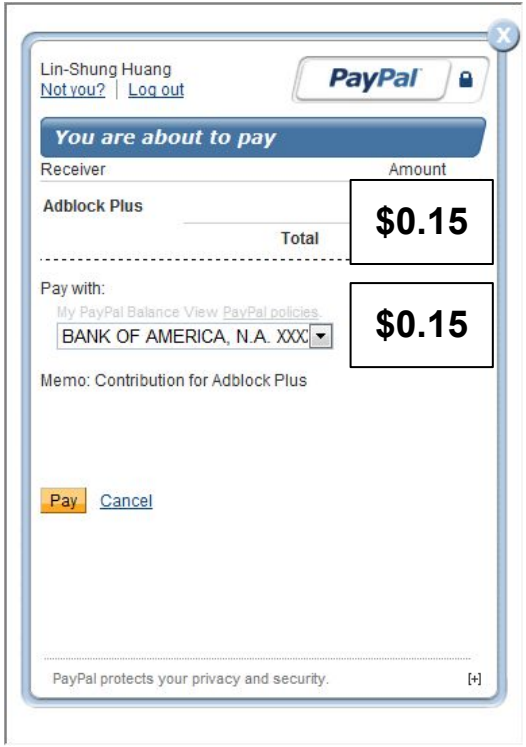
- Variant #2: Frame the legitimate site visibly, *under* invisible malicious content
  - Victim thinks they're clicking on the legitimate site
  - But their click actually happened on the malicious website!



# Clickjacking: Invisible iframes



- Variant #3: Frame the legitimate site visibly, *under* malicious content *partially* overlaying the site
  - The attacker can change the appearance of the site without breaking SOP!



# Clickjacking: Temporal Attack



- JavaScript can detect the position of the cursor and change the website right before the user clicks on something
  - The user clicks on the malicious input (embedded iframe, download button, etc.) before they notice that something changed

# Clickjacking: Temporal Attack



**Instructions:**
Please double-click on the button below to continue to your content

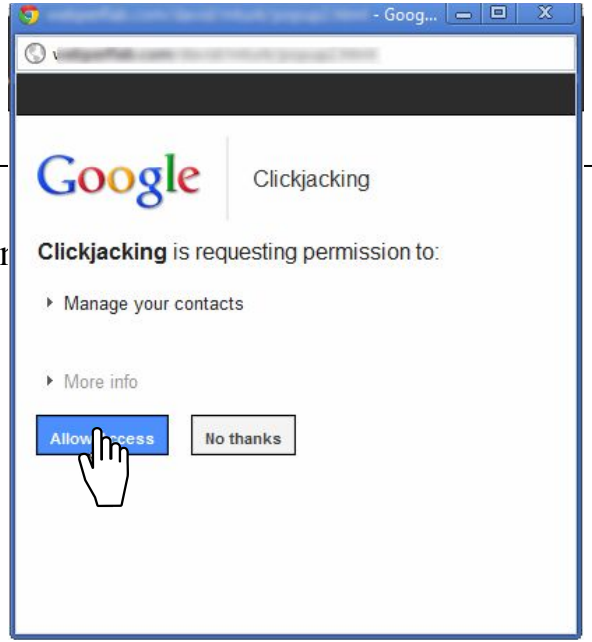


Click here

# Clickjacking: Temporal Attack



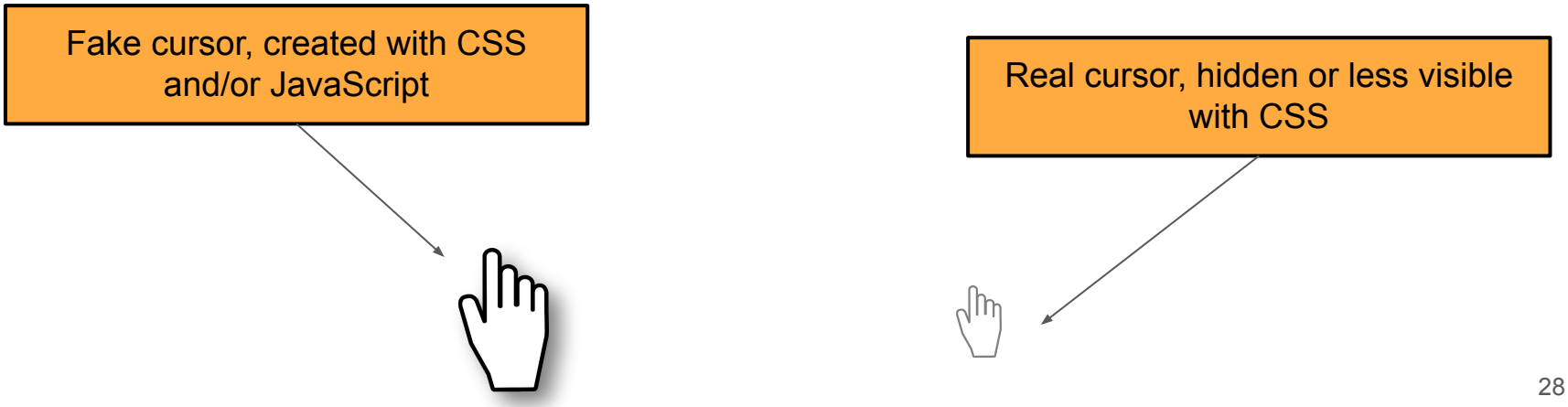
**Instructions:**
Please double-click on the buttor



# Clickjacking: Cursorjacking



- CSS has the ability to style the appearance of the cursor
- JavaScript has the ability to track a cursor's position
- If we change the appearance a certain way, we can create a fake cursor to trick users into clicking on things!



# Clickjacking: Cursorjacking



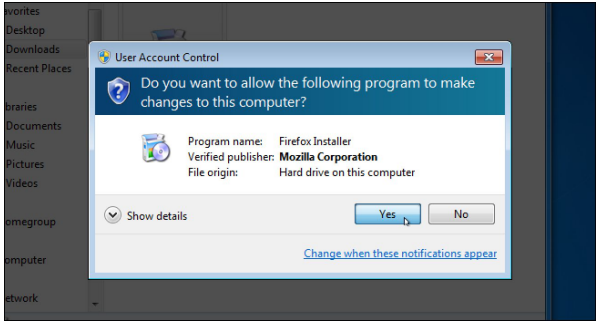
What do you think you're clicking on?

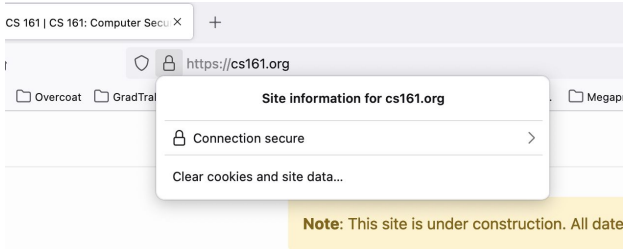


# Clickjacking: Defenses



- **Enforce visual integrity**: Ensure clear visual separation between important dialogs and content
  - Windows User Account Control darkens the entire screen and freezes the desktop
  - Firefox dialogs “cross the boundary” between the URL bar and content, something that only valid dialogs can do

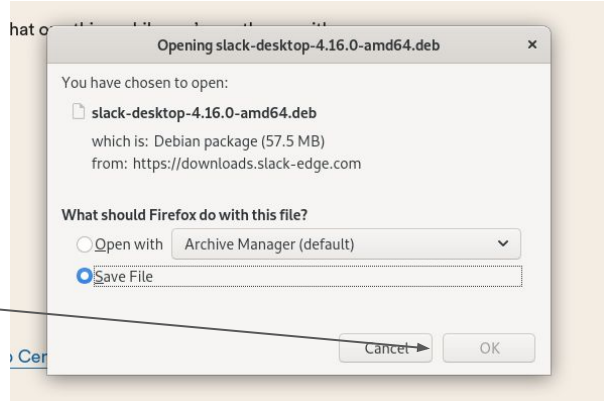


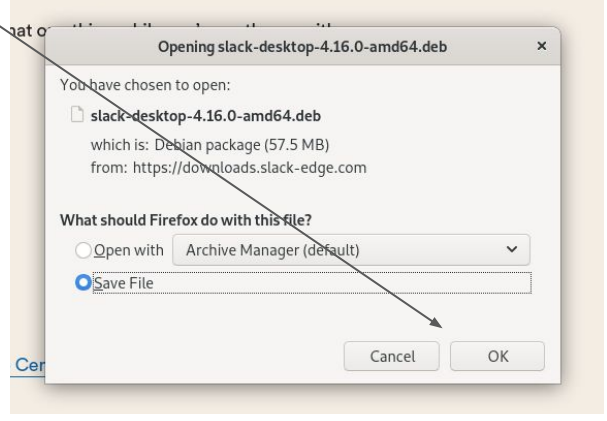



# Clickjacking: Defenses



- **Enforce temporal integrity**: Ensure that there is sufficient time for a user to register what they are clicking on
  - Notice: Firefox blocks the “OK” button until 1 second after the dialog has been focused



# Clickjacking: Defenses



- **Require confirmation** from users
  - The browser needs to confirm that the user's click was intentional
  - Drawbacks: Asking for confirmation annoys users (consider human factors!)
- **Frame-busting**: The legitimate website forbids other websites from embedding it in an iframe
  - Defeats the invisible iframe attacks
  - Can be enforced by Content Security Policy (CSP)
  - Can be enforced by X-Frame-Options (an HTTP header)

# Defense: CSP



- The CSP directive (sent as part of the HTTP response)

  **frame-ancestors** specifies which origins are allowed to embed the page in **<frame>**, **<iframe>**, **<object>**, or **<embed>** elements. If a site not listed tries to frame the page, the browser blocks it.

- Example:
  - Content-Security-Policy: frame-ancestors 'none';
    - No site can embed this page in a frame.
  - Content-Security-Policy: frame-ancestors 'self';
    - Only pages from the same origin can embed the page

# Defense: X-Frame-Options



- X-Frame-Options is an HTTP header, a legacy defense with modern web systems relying on CSP
- This header is included in the **HTTP response** from the web server

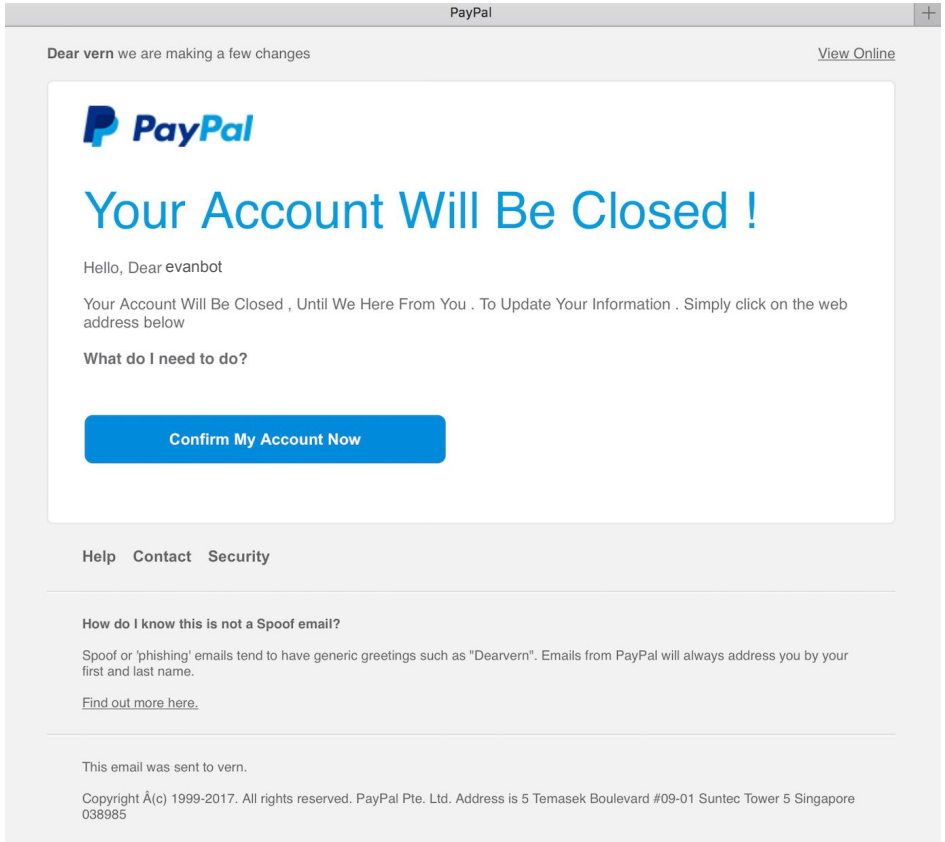
`X-Frame-Options: <value>`

Possible values:

- DENY: The page cannot be displayed in any frame, regardless of origin.
- SAMEORIGIN: Only pages from the same origin can frame the content.

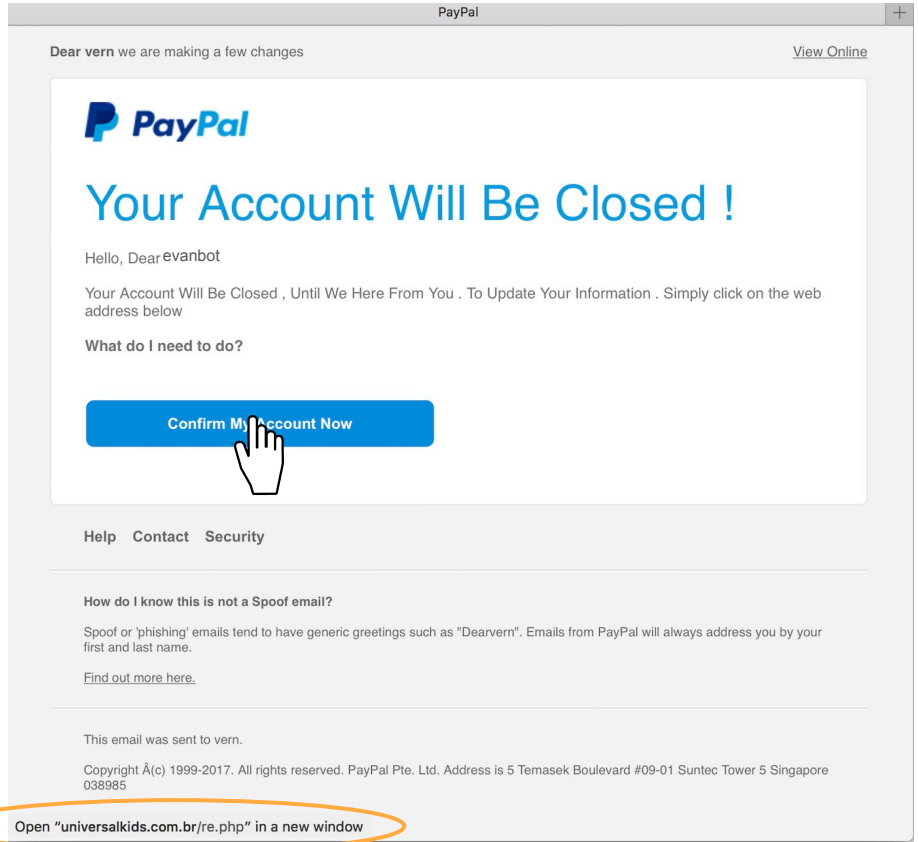
# Phishing




PayPal
Dear vern we are making a few changes
View Online
PayPal
Your Account Will Be Closed !
Hello, Dear evanbot
Your Account Will Be Closed , Until We Here From You . To Update Your Information . Simply click on the web address below
What do I need to do?
Confirm My Account Now
Help Contact Security
How do I know this is not a Spoof email?
Spoof or 'phishing' emails tend to have generic greetings such as "Dearvern". Emails from PayPal will always address you by your first and last name.
Find out more here.
This email was sent to vern.
Copyright Â(c) 1999-2017. All rights reserved. PayPal Pte. Ltd. Address is 5 Temasek Boulevard #09-01 Suntec Tower 5 Singapore 038985

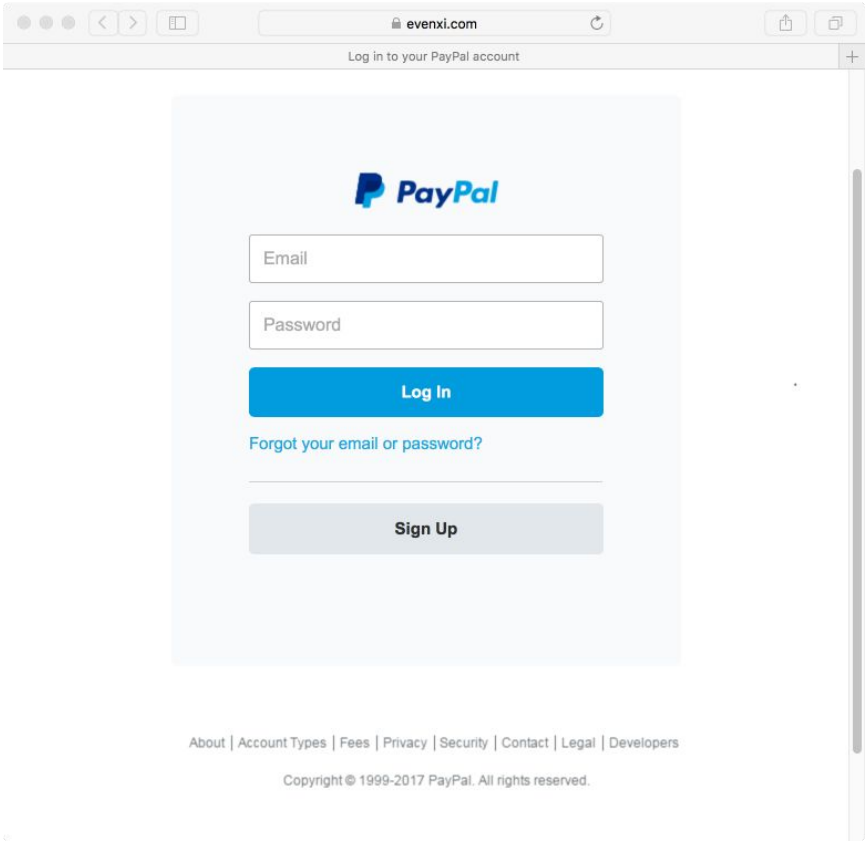
# Phishing




PayPal
Dear vern we are making a few changes
View Online
PayPal
Your Account Will Be Closed !
Hello, Dear evanbot
Your Account Will Be Closed , Until We Here From You . To Update Your Information . Simply click on the web address below
What do I need to do?
Confirm My Account Now
Help Contact Security
How do I know this is not a Spoof email?
Spoof or 'phishing' emails tend to have generic greetings such as "Dearvern". Emails from PayPal will always address you by your first and last name.
Find out more here.
This email was sent to vern.
Copyright Â(c) 1999-2017. All rights reserved. PayPal Pte. Ltd. Address is 5 Temasek Boulevard #09-01 Suntec Tower 5 Singapore 038985
Open "universalkids.com.br/re.php" in a new window

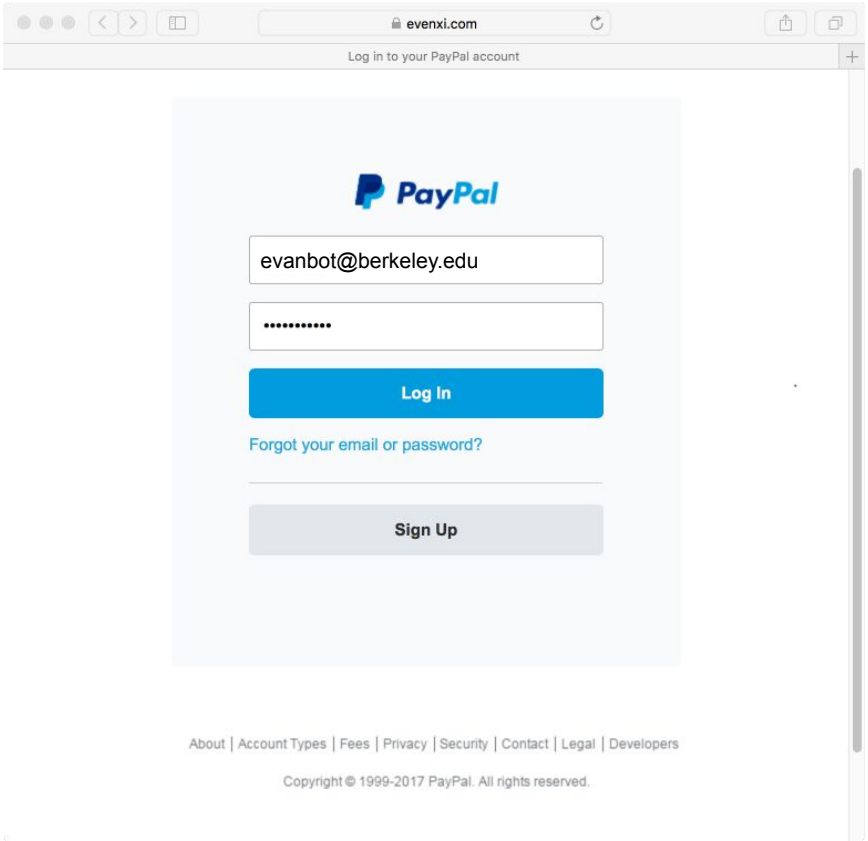
# Phishing




evenxi.com
Log in to your PayPal account
PayPal
Email
Password
Log In
Forgot your email or password?
Sign Up
About | Account Types | Fees | Privacy | Security | Contact | Legal | Developers
Copyright © 1999-2017 PayPal. All rights reserved.

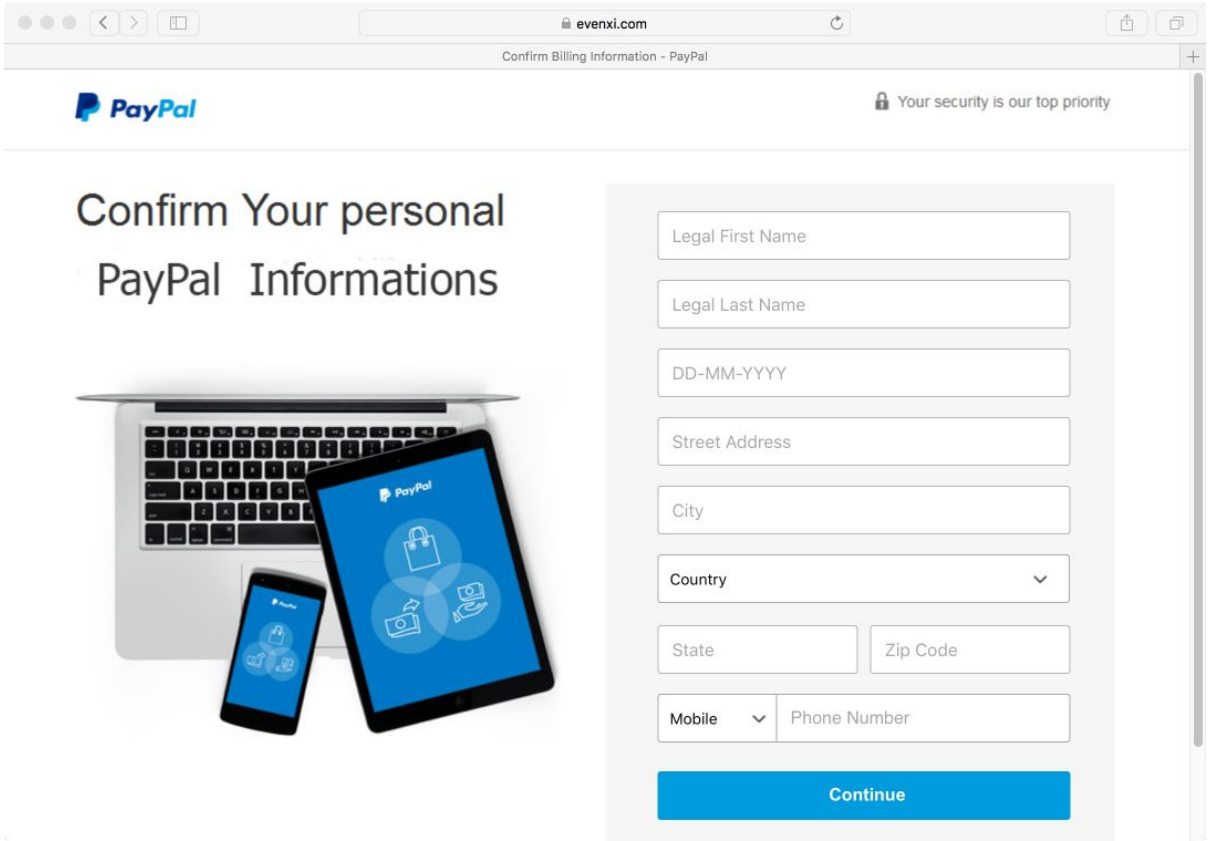
# Phishing




evenxi.com
Log in to your PayPal account
PayPal
evanbot@berkeley.edu
Log In
Forgot your email or password?
Sign Up
About | Account Types | Fees | Privacy | Security | Contact | Legal | Developers
Copyright © 1999-2017 PayPal. All rights reserved.

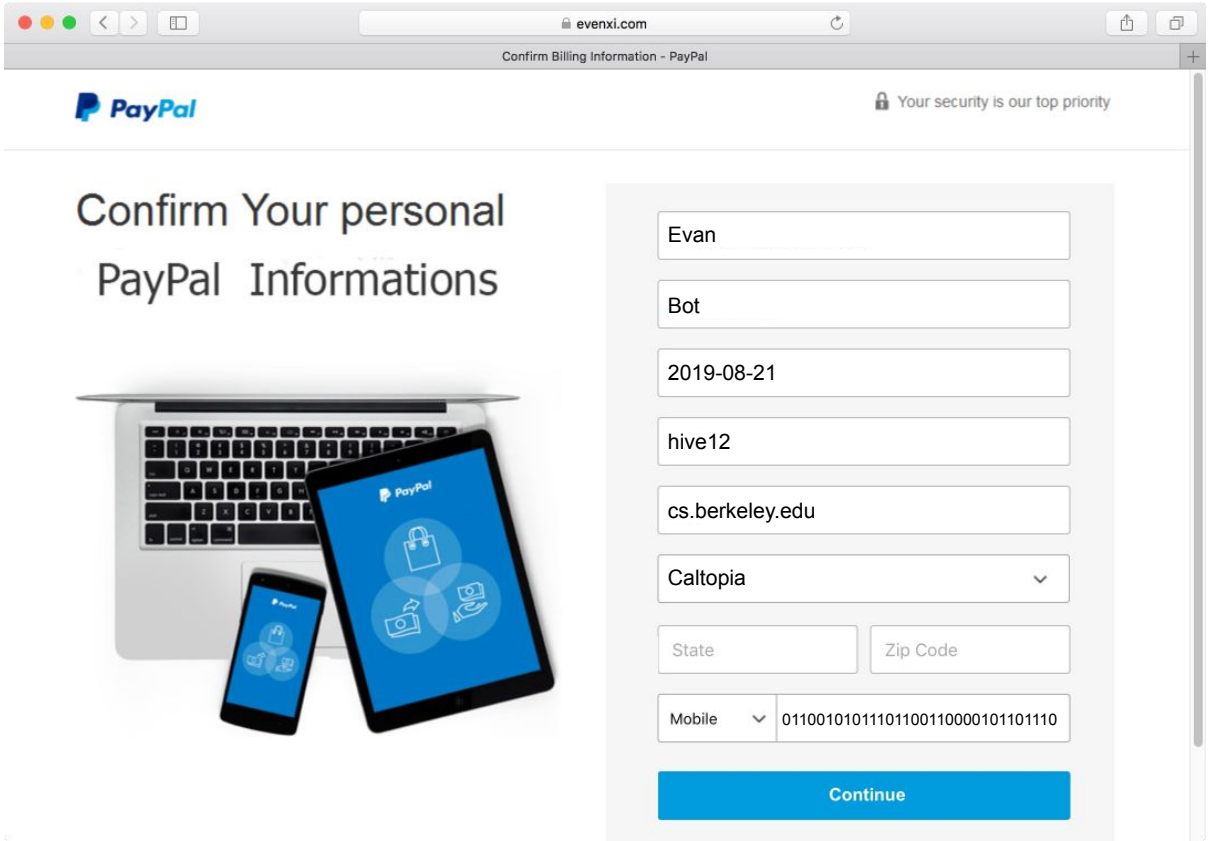
# Phishing




evenxi.com
Confirm Billing Information - PayPal
PayPal
Your security is our top priority
Confirm Your personal
PayPal Informations
Legal First Name
Legal Last Name
DD-MM-YYYY
Street Address
City
Country
State
Zip Code
Mobile
Phone Number
Continue

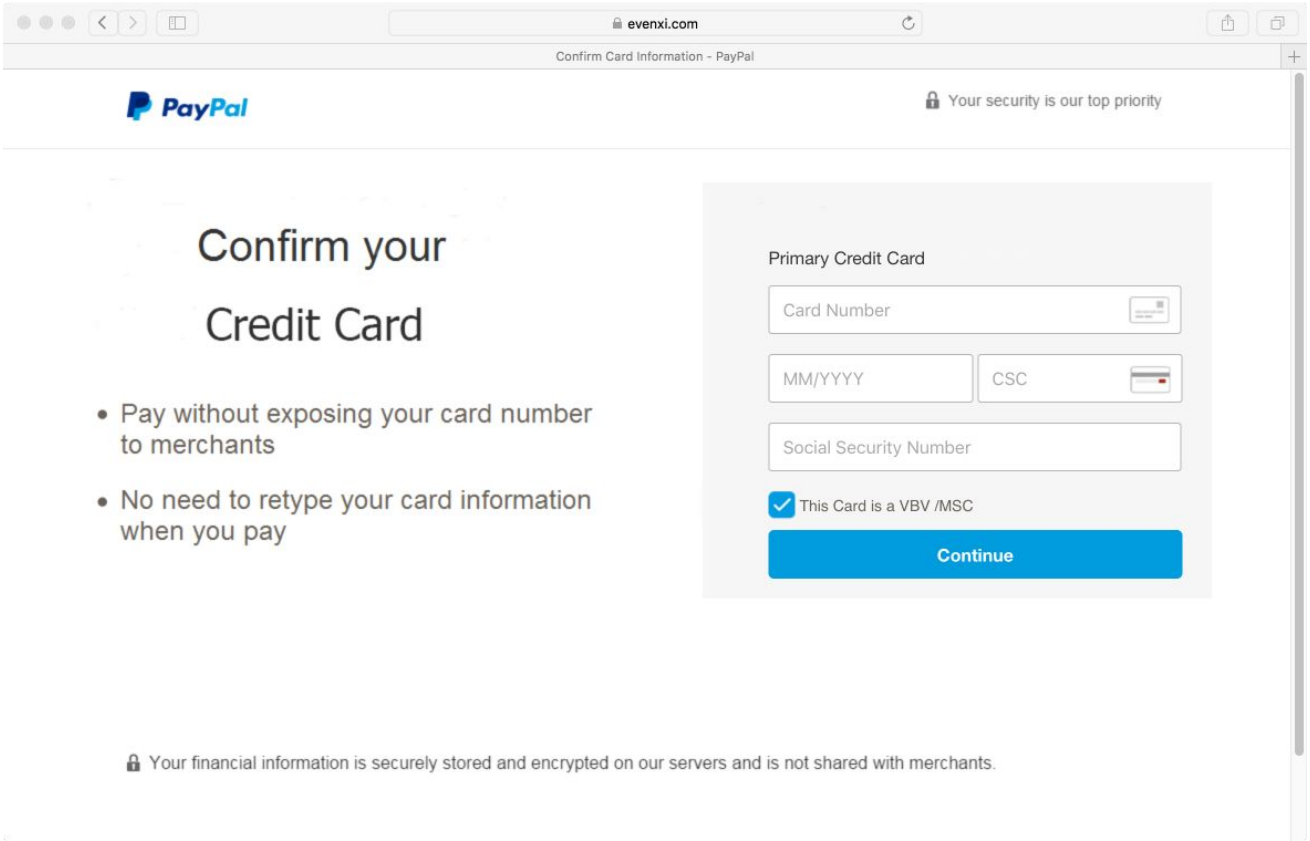
# Phishing




evenxi.com
Confirm Billing Information - PayPal
PayPal
Your security is our top priority
Confirm Your personal
PayPal Informations
Evan
Bot
2019-08-21
hive12
cs.berkeley.edu
Caltopia
State
Zip Code
Mobile
01100101011101100110000101101110
Continue

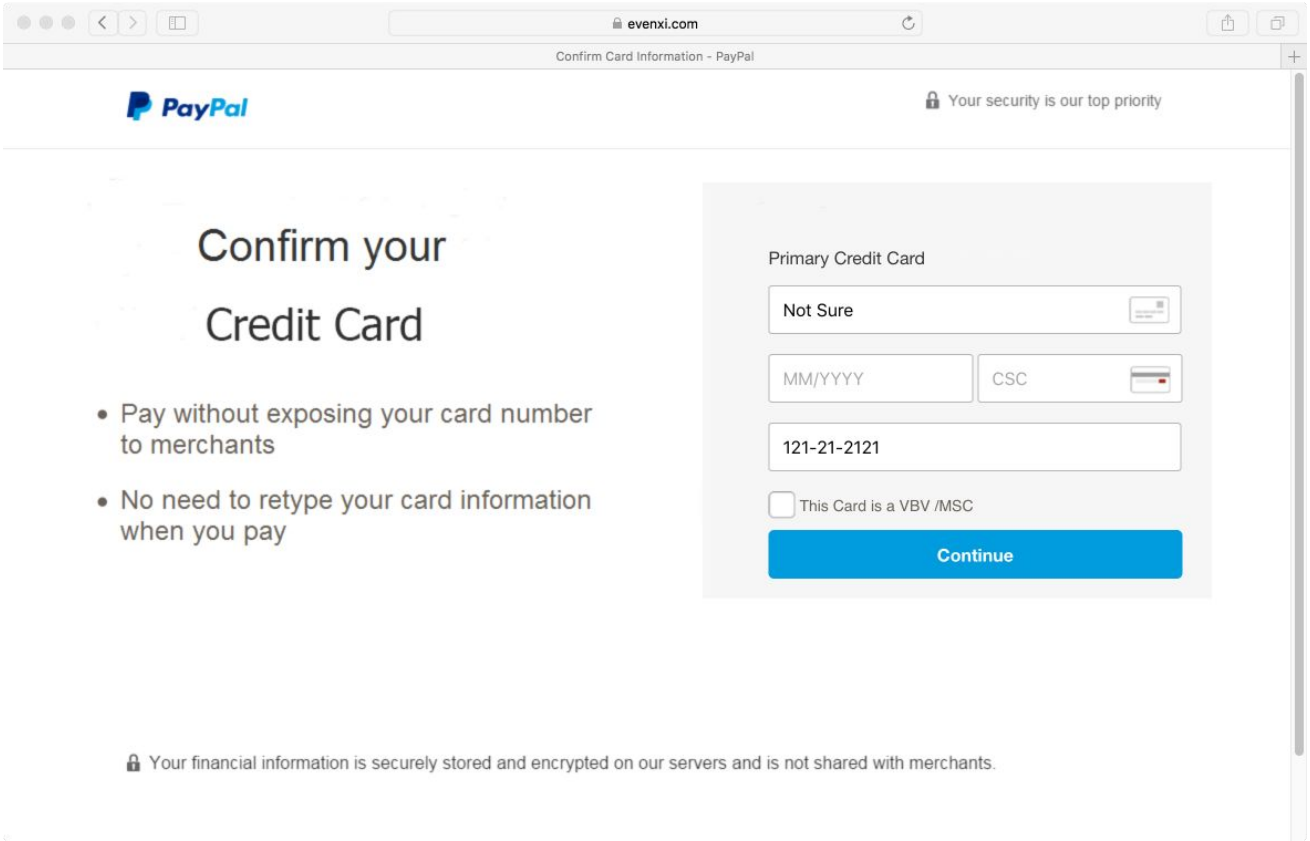
# Phishing




evenxi.com
Confirm Card Information - PayPal
PayPal
Your security is our top priority
Confirm your
Credit Card
• Pay without exposing your card number to merchants
• No need to retype your card information when you pay
Primary Credit Card
Card Number
MM/YYYY
CSC
Social Security Number
This Card is a VBV /MSC
Continue
Your financial information is securely stored and encrypted on our servers and is not shared with merchants.

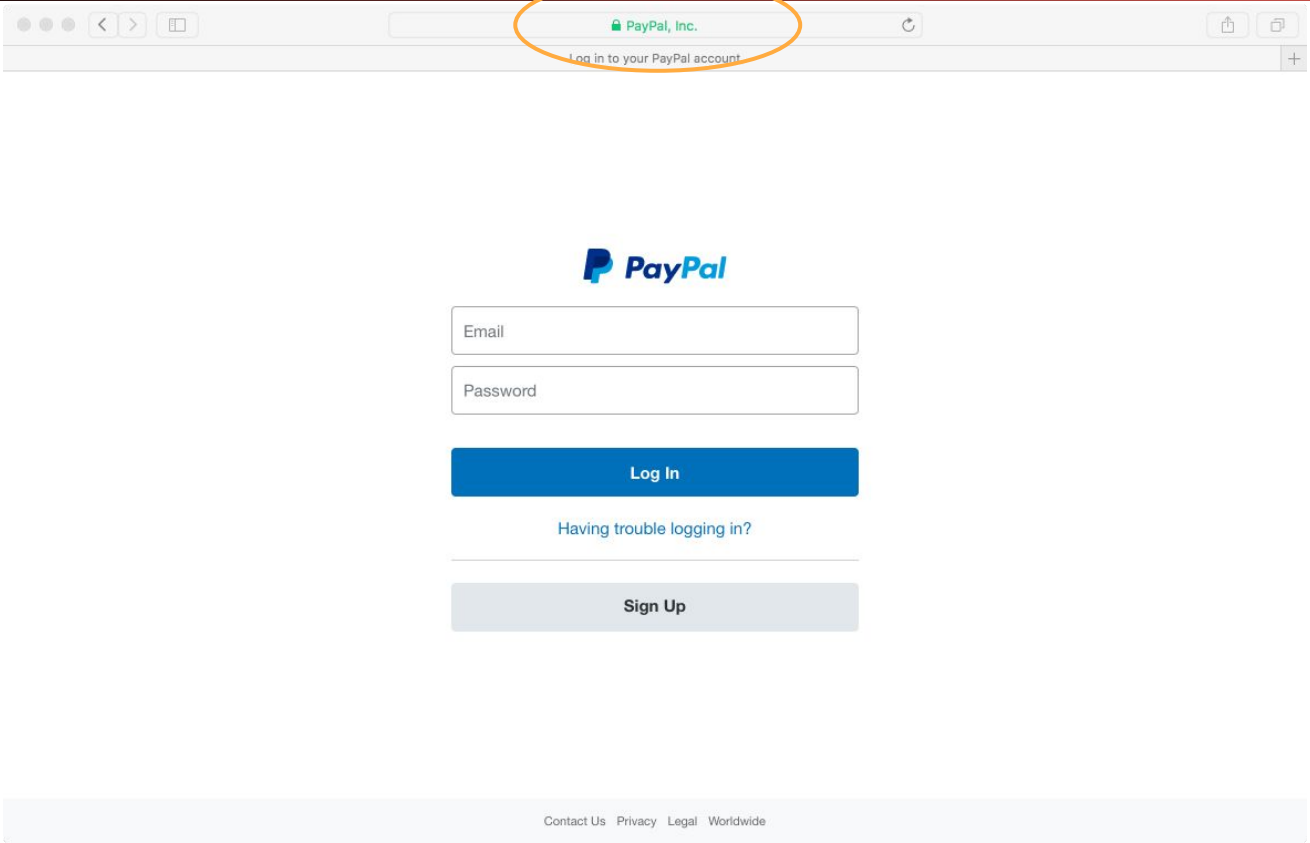
# Phishing




evenxi.com
Confirm Card Information - PayPal
PayPal
Your security is our top priority
Confirm your
Credit Card
• Pay without exposing your card number to merchants
• No need to retype your card information when you pay
Primary Credit Card
Not Sure
MM/YYYY
CSC
121-21-2121
This Card is a VBV /MSC
Continue
Your financial information is securely stored and encrypted on our servers and is not shared with merchants.

# Phishing




PayPal, Inc.
Log in to your PayPal account
PayPal
Email
Password
Log In
Having trouble logging in?
Sign Up
Contact Us Privacy Legal Worldwide

# Phishing

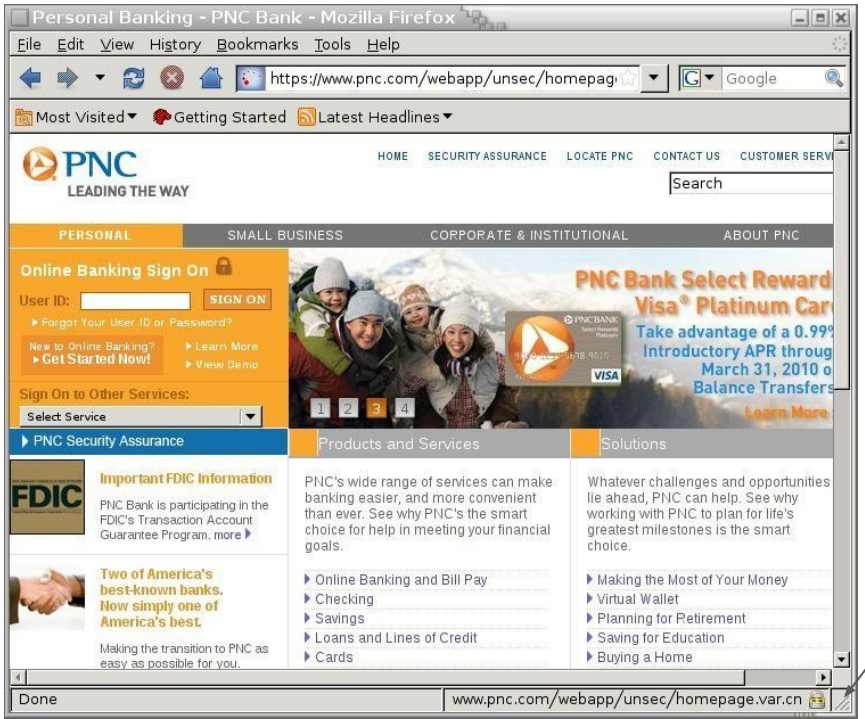


- **Phishing**: Trick the victim into sending the attacker personal information
- Main vulnerability: The user can’t distinguish between a legitimate website and a website *impersonating* the legitimate website

# Phishing: Check the URL?



Is this real?



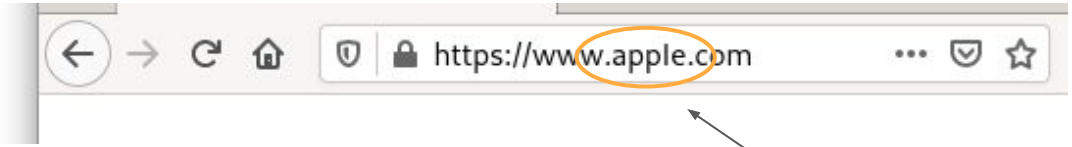

www.pnc.com/webapp/unsec/homepage.var.cn is actually an entire domain!

The attacker can still register an HTTPS certificate for the perfectly valid domain

# Phishing: Check the URL?



Is *this* real?




These letters come from the Cyrillic alphabet, not the Latin alphabet! They're rendered the same but have completely different bytes!

# Phishing: Homograph Attacks



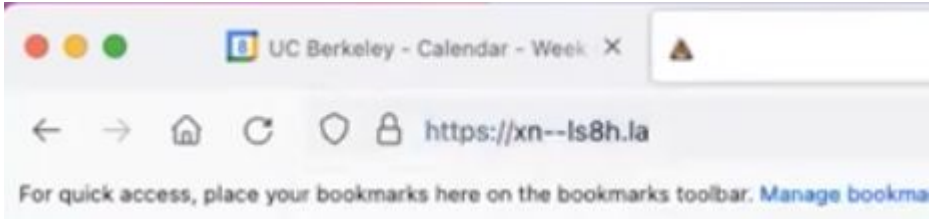
- Idea: Check if the URL is correct?
- **Homograph attack**: Creating malicious URLs that look similar (or the same) to legitimate URLs
  - Homograph: Two words that look the same, but have different meanings









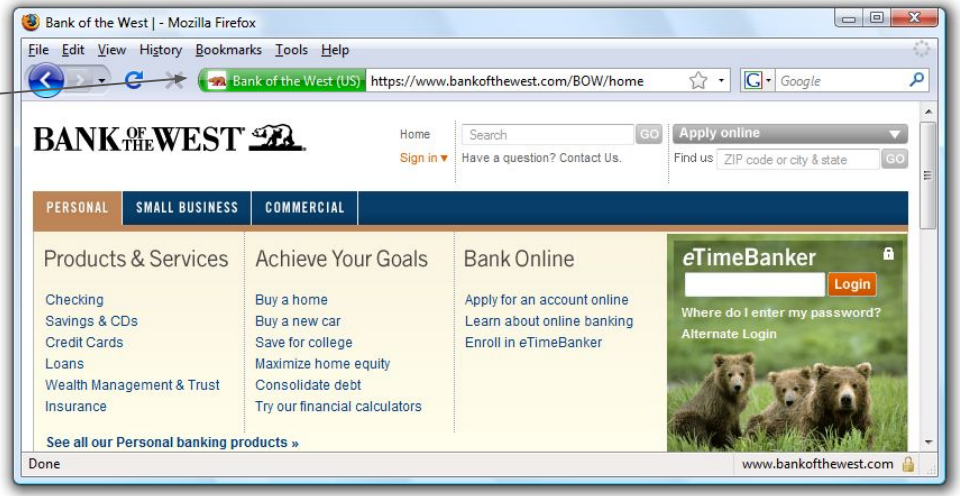


# Phishing: Check *Everything*



Is *this* real?

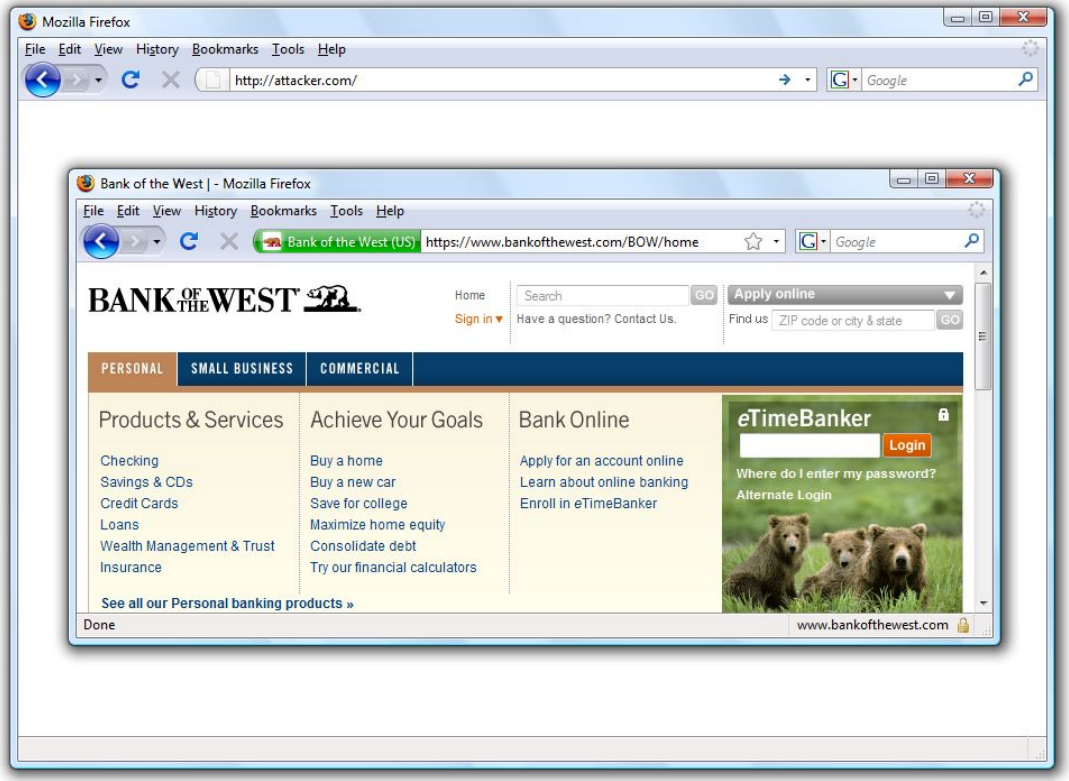
Extended Validation: Certificate authority verified the identity of the site (not just the domain)



# Phishing: Check *Everything*



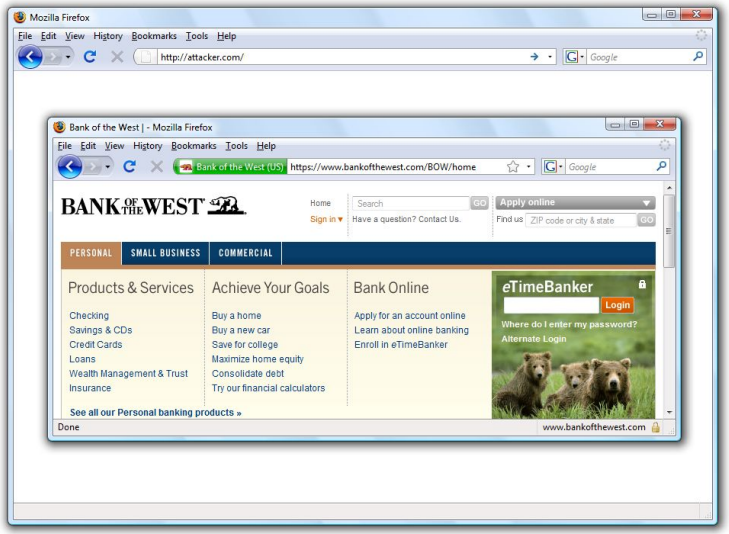
Oops, never mind



# Phishing: Browser-in-browser Attacks



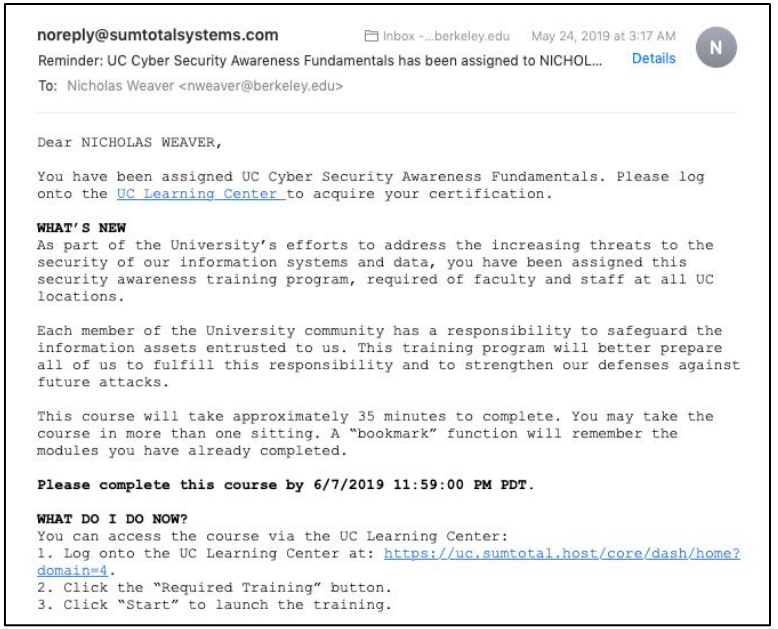
- Idea: Check for a green padlock icon in the browser's address bar, or any other built-in browser security feature
- **Browser-in-browser attack**: The attacker simulates the entire web browser with JavaScript



# Phishing: Don’t Blame the Users



- Most users aren’t security experts
- Attacks are uncommon: users normally don't suspect malicious action
- Detecting phishing is hard, even if you’re on the lookout for attacks
  - Legitimate messages often look like phishing attacks!



noreply@sumtotalsystems.com Inbox -...berkeley.edu May 24, 2019 at 3:17 AM

Reminder: UC Cyber Security Awareness Fundamentals has been assigned to NICHOL... Details

To: Nicholas Weaver <nweaver@berkeley.edu>

Dear NICHOLAS WEAVER,

You have been assigned UC Cyber Security Awareness Fundamentals. Please log onto the UC Learning Center to acquire your certification.

**WHAT'S NEW**
As part of the University's efforts to address the increasing threats to the security of our information systems and data, you have been assigned this security awareness training program, required of faculty and staff at all UC locations.

Each member of the University community has a responsibility to safeguard the information assets entrusted to us. This training program will better prepare all of us to fulfill this responsibility and to strengthen our defenses against future attacks.

This course will take approximately 35 minutes to complete. You may take the course in more than one sitting. A "bookmark" function will remember the modules you have already completed.

**Please complete this course by 6/7/2019 11:59:00 PM PDT.**

**WHAT DO I DO NOW?**
You can access the course via the UC Learning Center:
1. Log onto the UC Learning Center at: https://uc.sumtotal.host/core/dash/home?domain=4.
2. Click the "Required Training" button.
3. Click "Start" to launch the training.

# Two-Factor Authentication



- Problem: Phishing attacks allow attackers to learn passwords
- Idea: Require more than passwords to log in
- **Two-factor authentication** (**2FA**): The user must prove their identity in two different ways before successfully authenticating
- Three main ways for a user to prove their identity
  - **Something the user knows**: Password, security question (e.g. name of your first pet)
  - **Something the user has**: Their phone, their security key
  - **Something the user is**: Fingerprint, face ID
- Even if the attacker steals the user's password with phishing, they don't have the second factor

# Two-Factor Authentication



- Two-factor authentication also defends against other attacks where a user's password is compromised
  - Example: An attacker steals the password file and performs a dictionary attack
  - Example: The user reuses passwords on two different websites. The attacker compromises one website and tries the same password on the second website
  - With 2FA, the password alone is no longer enough for the attacker to log in

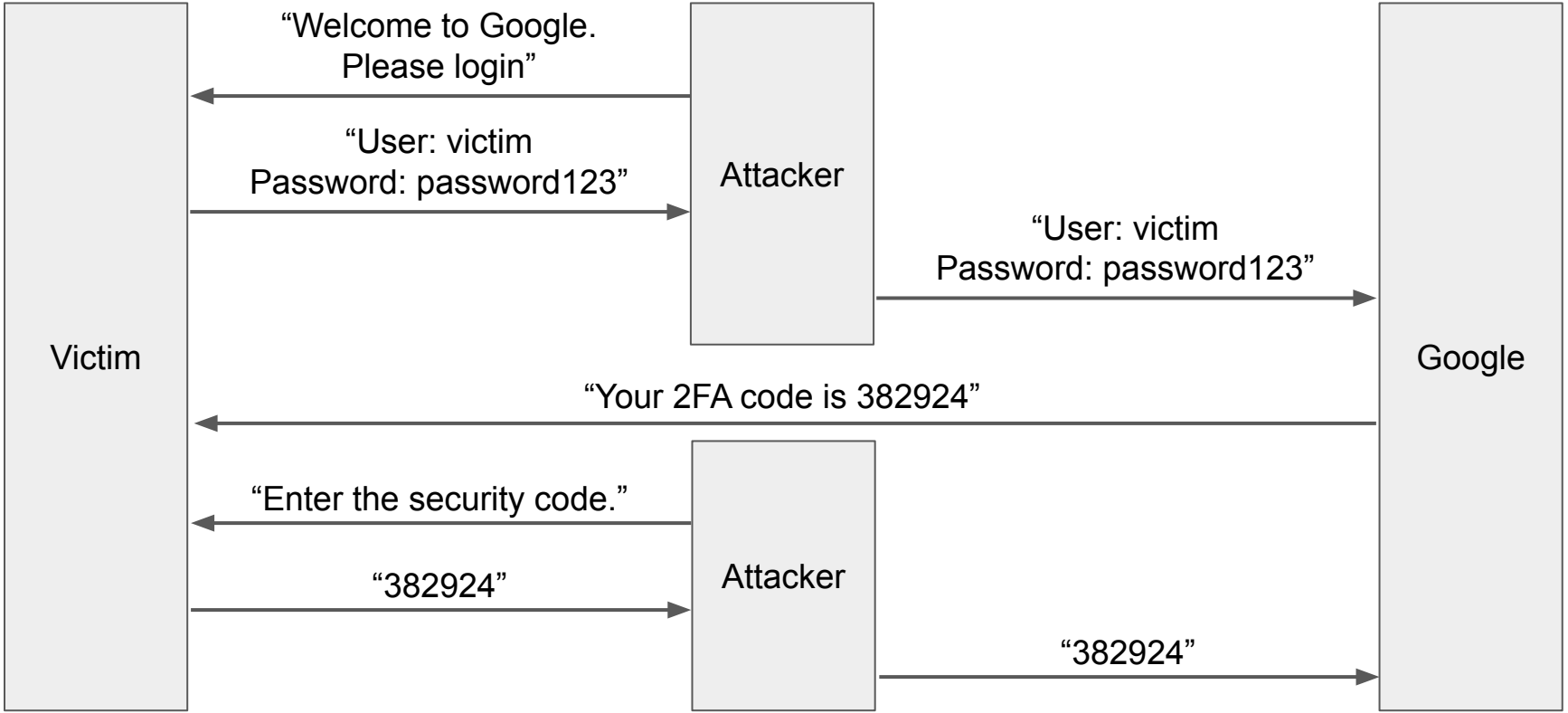
# Subverting 2FA: Relay Attacks



- **Relay attacks** (**transient phishing**): The attacker steals both factors in a phishing attack. Another form of man-in-the-middle attack.
- Example
  - Two-factor authentication scheme
    - First factor: The user's password (something the user knows)
    - Second factor: A code sent to the user's phone (something the user owns)
  - Attack
    - The phishing website asks the user to input their password (first factor)
    - The attacker immediately tries to log in to the actual website as the user
    - The actual website sends a code to the user
    - The phishing website asks the user to enter the code (second factor)
    - The attacker enters the code to log in as the user

# Subverting 2FA: Relay Attacks





# Subverting 2FA: Social Engineering



- Some 2FA schemes text a one-time code to a phone number
  - Attackers can call your phone provider (e.g., Verizon) and tell them to activate the attacker's SIM card, so they receive your texts!
  - 2FA via SMS is not great but *better than nothing* (use Authenticator app instead)
- Some 2FA schemes can be bypassed with customer support
  - Attackers can call customer support and ask them to deactivate 2FA!
  - Companies should validate identity if you ask to do this (but not all do)

# 2FA Example: Authentication Tokens



- **Authentication token**: A device that generates secure second-factor codes
  - Something the user owns
  - Examples: RSA SecurID and Google Authenticator
- Usage
  - The token and the server share a common secret key *k*
  - When the user wants to log in, the token generates a code HMAC(*k*, time)
    - The time is often truncated to the nearest 30 seconds for usability
    - The code is often truncated to 6 digits for usability
  - The user submits the code to the website
  - The website uses its secret key to regenerate the code and compare
- Drawback: Vulnerable to relay attacks
- Drawback: Vulnerable to online brute-force attacks
  - Possible fix: rate limits

# 2FA Example: Security Keys



- **Security key**: A device designed to defend against phishing
  - Something the user owns
  - Typically implement standards like FIDO2 and WebAuthn
- Usage
  - When the user signs up for a website, the security key generates a new public/private key pair and gives the public key to the website
  - When the user wants to log in, the server sends a nonce to the security key
  - The security key signs the nonce and website name (from the browser) and gives the signature to the server
- Security keys prevent phishing
  - Authentication is bound to the **origin (domain) that the browser sees**
  - In a phishing attack, the security key generates a signature with the attacker's website name, not the legitimate website name
    - Impervious to relay attacks!

# Summary: Clickjacking



- Clickjacking: Trick the victim into clicking on something from the attacker
- Main vulnerability: the browser trusts the user's clicks
  - When the user clicks on something, the browser assumes the user intended to click there
- Examples
  - Fake download buttons
  - Show the user one frame, when they're actually clicking on another invisible frame
  - Temporal attack: Change the cursor just before the user clicks
  - Cursorjacking: Create a fake mouse cursor with JavaScript
- Defenses
  - Enforce visual integrity: Focus the user's vision on the relevant part of the screen
  - Enforce temporal integrity: Give the user time to understand what they're clicking on
  - Ask the user for confirmation
  - Frame-busting: The legitimate website forbids other websites from embedding it in an iframe

# Summary: Phishing



- Phishing: Trick the victim into sending the attacker personal information
  - A malicious website impersonates a legitimate website to trick the user
- Don’t blame the users
  - Detecting phishing is hard, especially if you aren’t a security expert
  - Check the URL? Still vulnerable to homograph attacks (malicious URLs that look legitimate)
  - Check the entire browser? Still vulnerable to browser-in-browser attacks
- Defense: Two-Factor Authentication (2FA)
  - User must prove their identity two different ways (something you know, something you own, something you are)
  - Defends against attacks where an attacker has only stolen one factor (e.g. the password)
  - Vulnerable to relay attacks: The attacker phishes the victim into giving up both factors
  - Vulnerable to social engineering attacks: Trick humans to subvert 2FA
  - Example: Authentication tokens for generating secure two-factor codes
  - Example: Security keys to prevent phishing